Code No: 117CD **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, September - 2021
DATA WAREHOUSING AND DATA MINING
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Answer any Five Questions
All Questions Carry Equal Marks
- - -

1.a) Define Data warehouse. Discuss Construction of Multi-dimensional model and its operations.
b) Explain the fact constellation schema with an example. [7+8]

2. Explain the various stages of Data preprocessing.
Explain Binning – techniques and Data - normalization. Let the values for the attribute age are
{ 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70.} Smooth this data by bin means using a bin depth of 3. [15]

3.a) Explain the Data mining architecture.
b) Given two objects represented by the tuples (22,1,42,10) and (20,0,36,8)
i) Compute the Euclidean distance between the two objects.
ii) Compute the Manhattan distance between the two objects.
iii) Compute the Minkowski distance between the two objects using q=3;
iv) Compute Jaccard distance between the two objects. [7+8]

4. Explain FP-Growth algorithm and generate effective association rules with minimum support count =33%. [15]

Transaction	LIST OF ITEMS
001	MILK, DAL, SUGAR, BREAD
002	DAL, SUGAR, WHEAT,JAM
003	MILK, BREAD, CURD, PANEER
004	WHEAT, PANEER, DAL, SUGAR
005	MILK, PANEER, BREAD
006	WHEAT, DAL, PANEER, BREAD

5.a) Compare and contrast Eager classifiers with Lazy classifiers What are the various methods of evaluating accuracy of a classifier or predictor? Explain Bagging and Boosting techniques?
b) Explain Naïve-Bayer classified. [7+8]

6. Perform KNN classification for the data given below for X= {P1= 3,P2=7} where k=3. [15]

P1	P2	CLASS
7	7	FALSE
7	4	FALSE
3	4	TRUE
1	4	TRUE

7. Define Outlier Analysis. Explain the various algorithms for identifying the outliers in a given cluster. Illustrate. [15]

8.a) Categorize major clustering methods.
b) Explain Hiererchical Clustering. [7+8]

---ooOoo---